

FactumDB User Guide

Installation, Evidence Examination, and Report Export

Application version 0.1.0

For digital forensic investigators

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1. Introduction and scope

FactumDB is a local desktop application for examining MySQL/InnoDB database evidence. It reconstructs record changes from row binary logs and compares the resulting log-derived state with physical records extracted from .ibd tablespace files.

The application is designed to help an examiner identify where evidence sources agree, where comparable observations conflict, and where the available material is insufficient. Findings retain links to supporting evidence and extraction outputs so that an examiner can trace how a conclusion was produced.

Interpretation boundary FactumDB assists examination; it does not automatically establish intent, authenticity, or completeness. A successful run means the configured processing completed. A conflict means observations disagree under the applicable rules. Both still require examiner interpretation.

What you need for a first examination

- A supported FactumDB desktop installation and a local examiner account.
- Relevant available InnoDB tablespace files (.ibd).
- The actual binary-log sequence needed for the period under examination, plus a binary-log index file when available.
- Enough local disk space for verified working copies and analysis results.

Evidence model in one sentence

Binary logs describe changes over time; .ibd tablespaces provide physical row observations. FactumDB correlates these sources and then reconciles log-derived state with physical evidence.

2. Supported platforms

The screenshots in this guide show the Windows build. The examination workflow is shared by Windows and Ubuntu after sign-in, although native dialogs, file pickers, and paths differ.

Platform	Verified target	Native sign-up verification	Notes
Windows	Windows x64	Windows Hello / Windows Security	Installer can obtain Microsoft WebView2 if it is missing.
Ubuntu	Ubuntu 24.04 x86-64/amd64 with graphical desktop	polkit authentication agent	Earlier Ubuntu versions, ARM systems, other Linux distributions, terminal-only environments, and WSL are not verified interactive targets.

3. System requirements

Component	Minimum	Recommended
Operating system	Ubuntu 24.04 (64-bit) or Windows (64-bit)	Ubuntu 24.04 or Windows (64-bit)
Processor	64-bit processor with 2 cores or CPU threads	Modern 64-bit processor with 2 or more cores
Memory (RAM)	4 GB	8 GB for additional headroom
Free storage	5 GB before installation, plus space for case data	10 GB or more, plus space for case data, exports, and backups

4. Install on Windows

Supported package target: Windows x64.

1. Obtain the supplied FactumDB NSIS setup .exe from the project distributor. Verify that the installer came from the expected source before running it.
2. Open the setup .exe and complete the installer. Exact wizard pages are not documented in this guide because no installation screenshots are supplied.
3. If Microsoft WebView2 is missing, the installer may need an internet connection to obtain it.
4. Launch FactumDB from the Windows Start menu.
5. Before creating the first examiner account, make sure Windows Hello is configured for the current Windows user.

Unsigned installer The current Windows installer is unsigned, so publisher verification may be unavailable. Confirm the installer source rather than routinely bypassing Windows security warnings.

The Windows package includes Python, extraction utilities, PDF dependencies, and the required Visual C++ runtime DLLs. MySQL is not installed and is not required for evidence examination.

5. Install on Ubuntu 24.04

Verified target: Ubuntu 24.04 x86-64/amd64 with a graphical desktop session.

1. Obtain the supplied .deb installer and open a terminal in the folder that contains it.
2. Update the local APT package index, then install the local package with APT.
3. Launch FactumDB from the desktop application menu, or run factumdb from a terminal as the normal desktop user.

```
sudo apt update
sudo apt install ./FactumDB_0.1.0_amd64.deb

# Launch
 factumdb
```

The filename must match the package you actually downloaded. The ./ prefix means “use the package file in the current directory.” If the version changes, use the matching filename.

Native verification requirement Initial sign-up needs an active graphical session with a working polkit authentication agent. GNOME and KDE normally provide one. A terminal-only session or WSL environment may not provide the required authentication prompt.

APT installs declared Python and system-library dependencies. The FactumDB package supplies extraction utilities, PDF libraries, a desktop launcher, and the FactumDB polkit sign-up policy. The sudo command is for installation only; launch FactumDB as the normal desktop user.

6. Create an examiner account and sign in

At first launch, the application presents the sign-in screen. Select “Create an account” to open the examiner account form shown in Figure 1.

1. Enter a username containing 3-64 ASCII letters, numbers, dots, underscores, or hyphens.
2. Choose a FactumDB password with at least 12 characters.
3. Enter the same password in “Confirm password”.
4. Select “Verify device and sign up”.
5. Complete the operating system verification prompt. The FactumDB account is created only after native verification succeeds.

Two different credentials The FactumDB password is the local application credential you choose. The operating-system prompt verifies the current device user. On Ubuntu this is commonly the Linux login password; on Windows it uses an available Windows Hello method.

The username “john” visible in later screenshots is demonstration data only. Use your own examiner account identity.

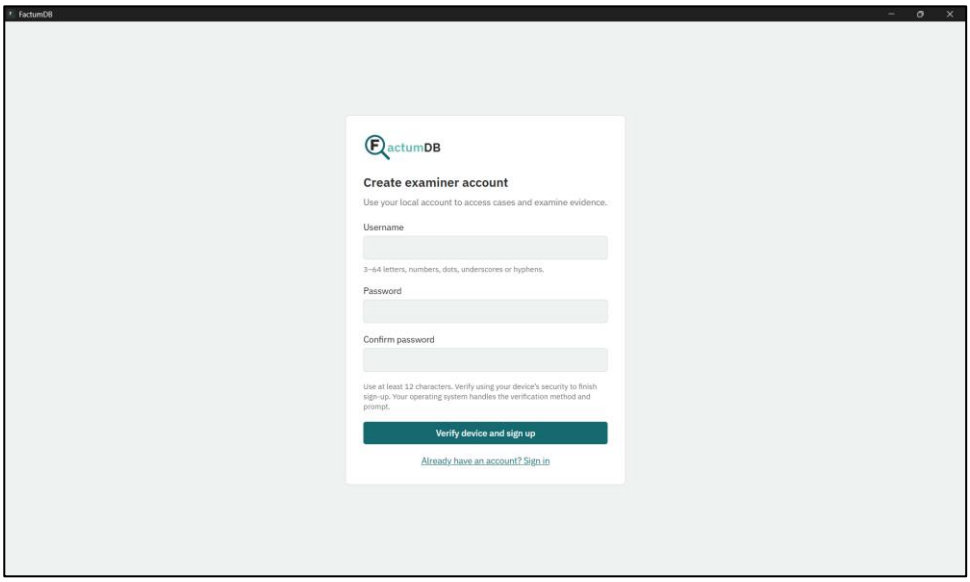


Figure 1. Create examiner account. The form enforces the local username format, password confirmation, and device verification step.

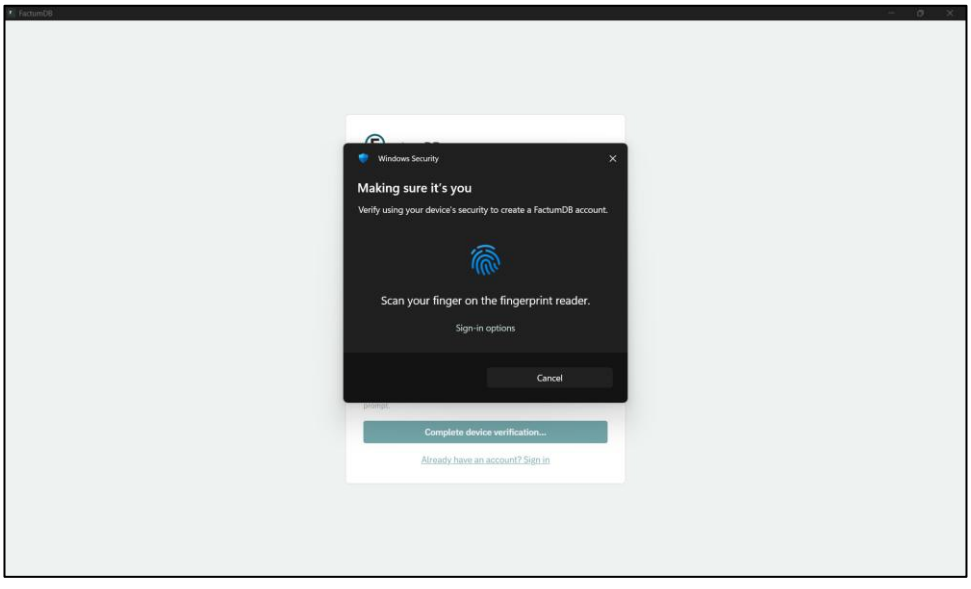


Figure 2. Windows Security verification during sign-up. Fingerprint is only an example; the available Windows Hello method depends on the device and user configuration.

Windows verification

Complete the Windows Security / Windows Hello challenge offered by the operating system. Do not assume a fingerprint reader is required; a configured PIN, biometric method, or another Windows Hello option may be offered.

Ubuntu verification

FactumDB uses the same account form, but Ubuntu presents a polkit authentication dialog rather than Windows Security. Authenticate the current Linux desktop account using the method offered by the system. This operating-system verification is separate from the FactumDB password you just chose.

Subsequent sign-in, Lock, and Sign out

Later sign-ins use the local FactumDB username and password. Cases are scoped to the signed-in examiner. The top bar provides “Lock” and “Sign out”. Unlocking requires the current FactumDB password. If an analysis stage is already executing when the interface is locked, that stage may finish.

If native verification fails Cancelling, timing out, or lacking the native authentication service prevents sign-up. On Windows, check Windows Hello. On Ubuntu, check the graphical desktop session, polkit authentication agent, and installed FactumDB policy. Do not bypass native verification.

7. Case management

1. Open the “Cases” tab.
2. Enter a meaningful case name and select “Create case”. The signed-in examiner is associated with the case.
3. Use “Refresh cases” to reload the list when needed. Use the filter field to narrow displayed cases.
4. To continue previous work, select an existing case from the case list rather than creating a duplicate case.
5. Open “Evidence” to review the case ID, case name, examiner, case workspace, and examiner notes. Use “Save notes” when you change the notes field.

Case identity Do not edit the examiner field to impersonate another examiner. Case ownership is associated with the signed-in local account. The example case name “test”, case ID, workspace path, and username in the screenshots are demonstration values.

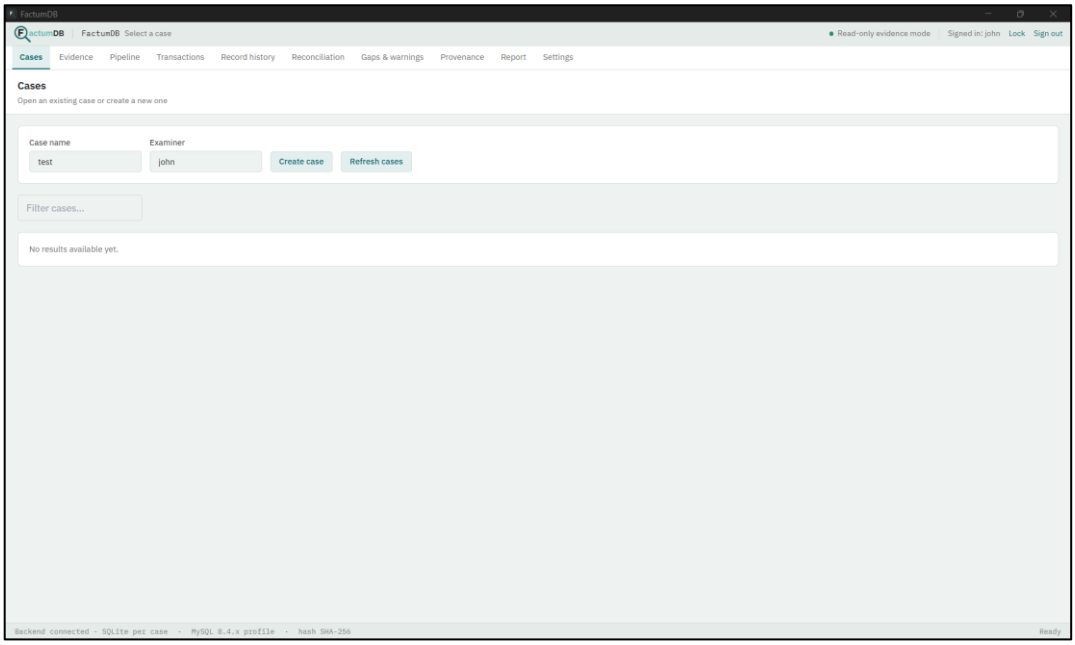


Figure 3. Cases screen. Create a named case, refresh the case list, filter cases, or reopen existing work.

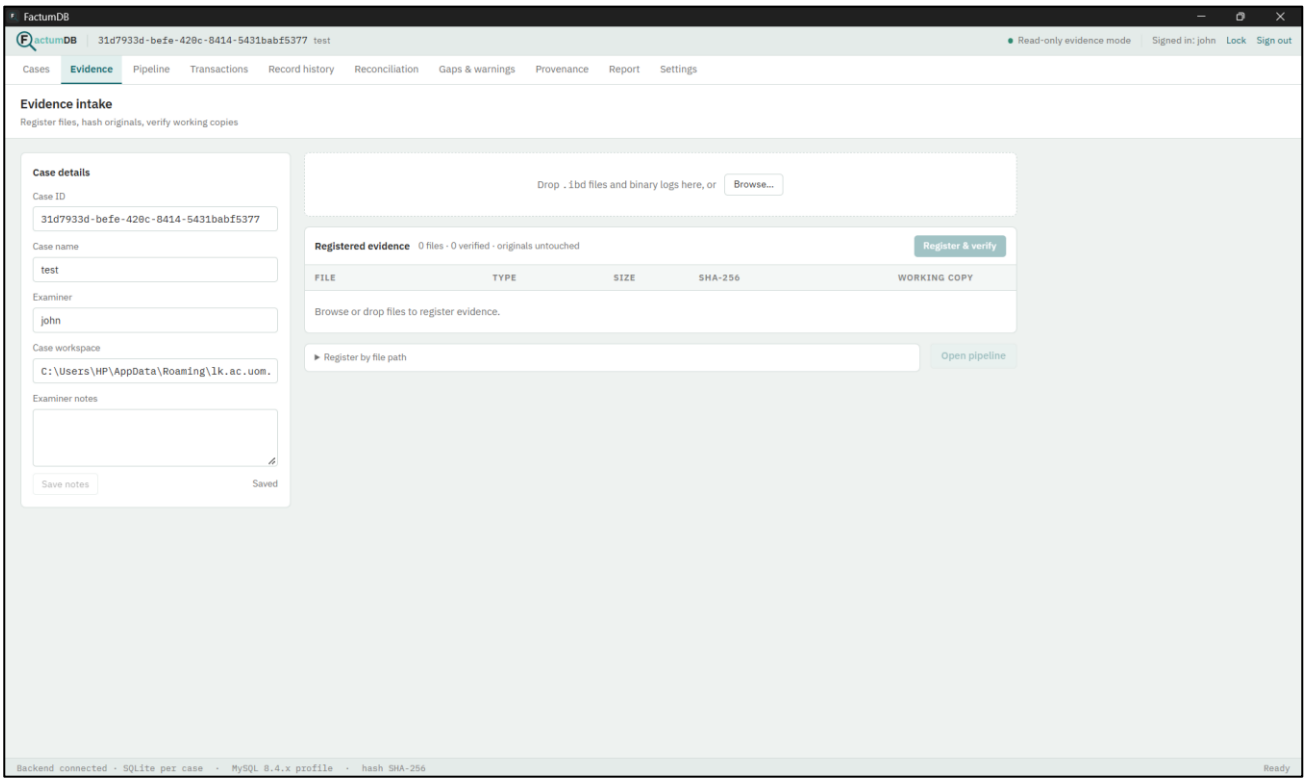


Figure 4. Evidence screen immediately after case creation. Case details and notes appear at left; the evidence intake area is ready for files.

8. Evidence intake and integrity

Supported evidence categories shown by FactumDB

- InnoDB tablespace files (.ibd).
- Row binary logs such as mysql-bin.000001.
- Binary-log index files such as mysql-bin.index. The index is supporting metadata and does not replace the actual binary logs.

1. Use “Browse...” or drag evidence files into the drop area. “Register by file path” is an alternate intake method.
2. Review the registered evidence list. FactumDB records SHA-256 hashes when files are registered.
3. Select “Register & verify” to verify the registered working copies.
4. Confirm that the Working Copy status changes from “Registered” to “Verified” before starting analysis.
5. Import the relevant available .ibd files and the actual binary-log sequence needed for the period under examination. Missing evidence can limit what can be reconstructed or compared.

> **Integrity meaning** A matching SHA-256 supports integrity verification between the registered evidence and the working copy used for processing. A hash match does not prove that the evidence is authentic, complete, or correctly acquired.

FactumDB preserves originals and processes verified working copies. The automatically assigned case workspace should normally be left as assigned.

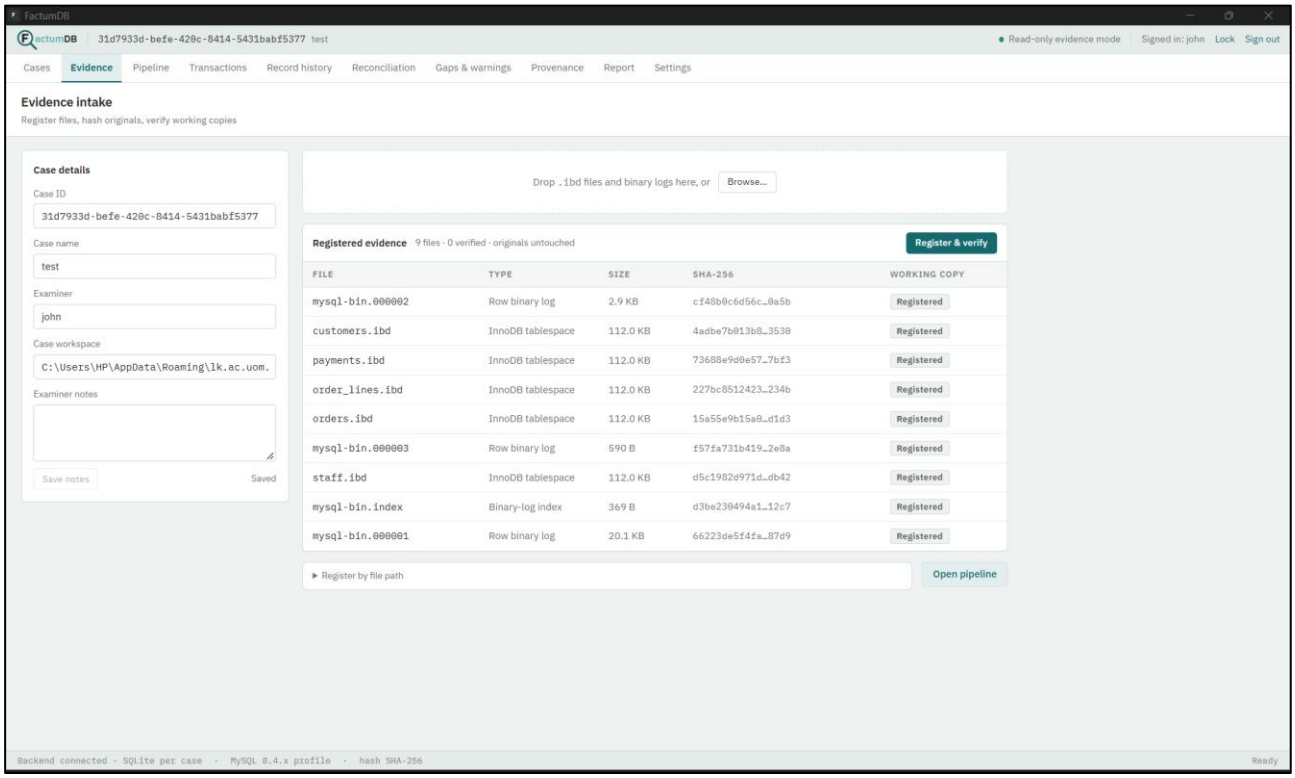


Figure 5. Evidence files registered but not yet verified. The list includes .ibd tablespaces, row binary logs, and a binary-log index.

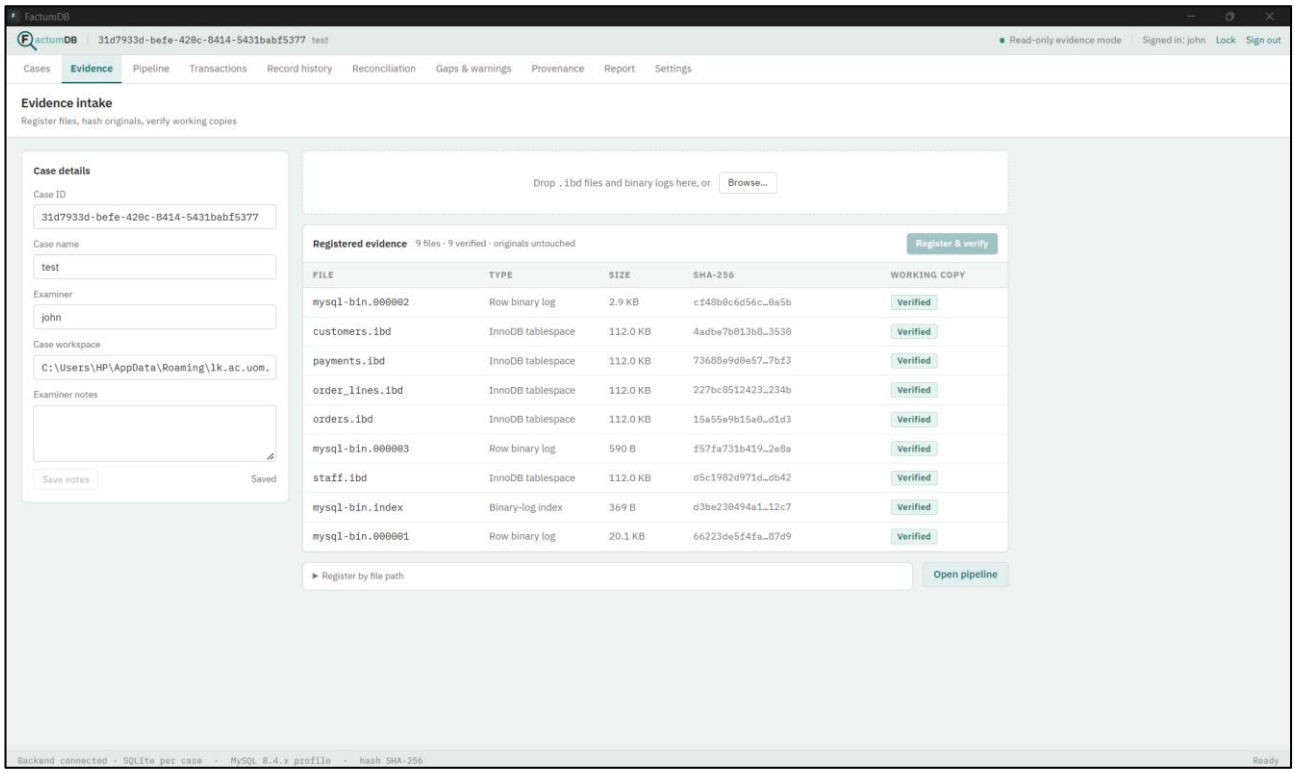


Figure 6. Working copies after verification. Each listed item now shows “Verified”.

9. Processing pipeline

Open “Pipeline” from the navigation bar or select “Open pipeline” from Evidence. FactumDB can run the full sequence automatically or advance one stage at a time.

Controls

- “Run complete analysis” executes the analysis sequence.
- “Run stage” advances processing stage by stage.
- “Cancel at stage boundary” requests cancellation between stages; it is not an immediate interruption of the current stage.
- A failed stage may expose “Retry failed stage”.
- After a stopped or completed run, “Start new analysis” is available. Do not restart a completed examination unless there is a reason to do so.

Workflow rows shown in the application

01 Evidence registration	08 Normalization
02 SHA-256 hashing	09 Transaction grouping
03 Verified working copies	10 Record correlation
04 InnoDB page validation	11 State reconstruction
05 Schema extraction	12 Reconciliation
06 Physical-row extraction	13 Result persistence & report
07 Binary-log decoding	

The progress indicator counts 10 analysis stages, corresponding to workflow rows 03-12. Rows 01-02 represent evidence registration and hashing, while row 13 persists results and prepares reporting. Therefore “10/10 analysis stages completed” is compatible with 13 displayed workflow rows.

Reading the dashboard Pending means the stage has not yet run. Running indicates the active stage. Completed rows show a result summary and may show duration. A “?” in a summary card means the result is not yet available; it does not mean zero.

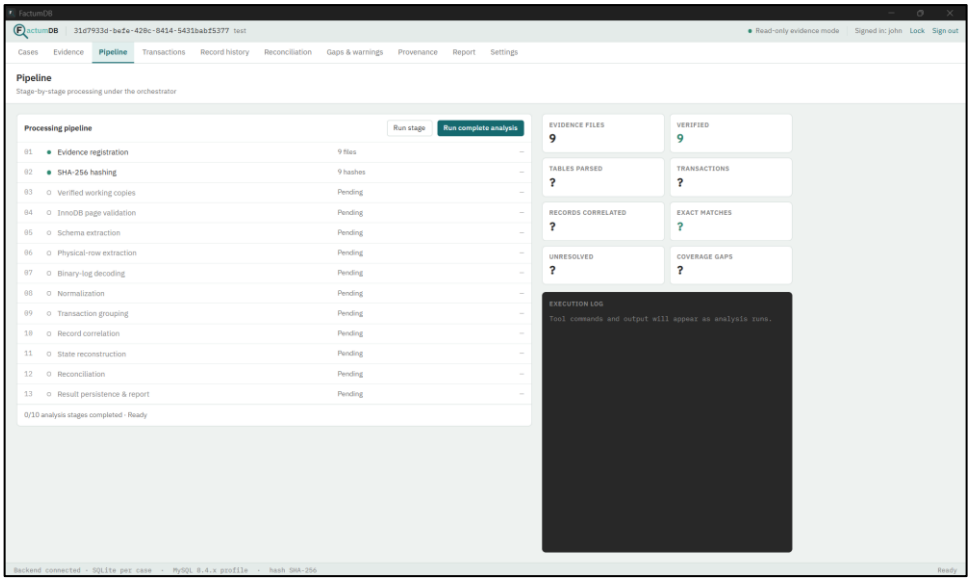


Figure 7. Pipeline before analysis. Evidence registration and hashing are present; analysis stages are pending and summary cards show “?” where results do not yet exist.

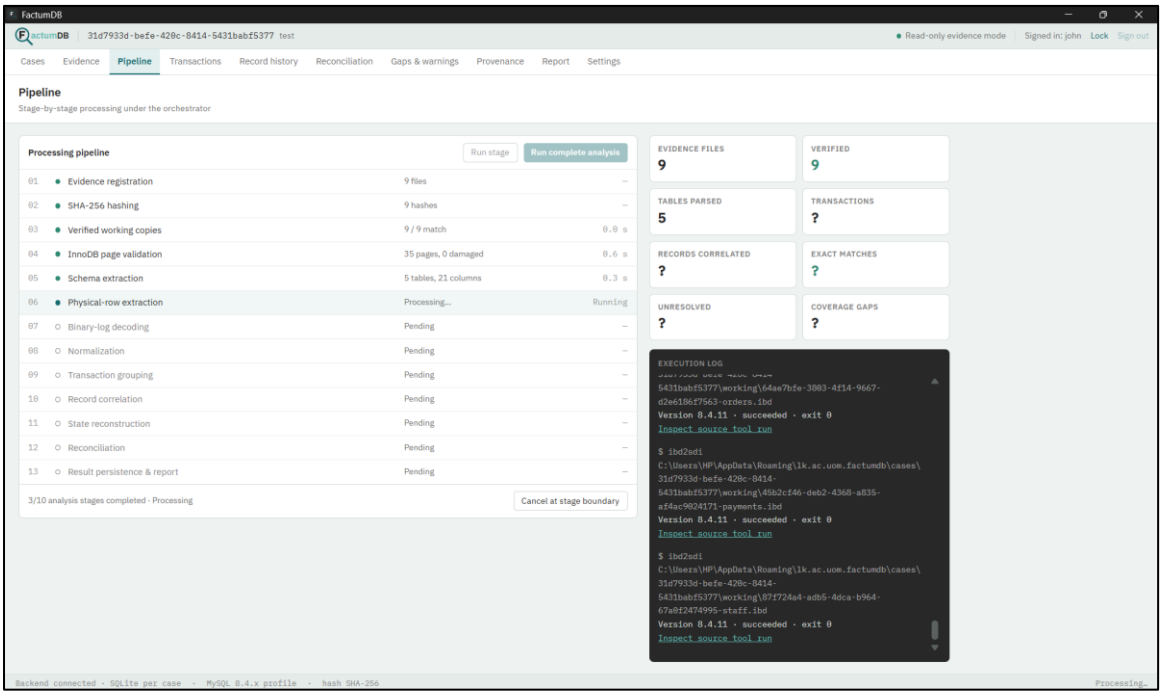


Figure 8. Analysis in progress. Completed stages show summaries and durations while Physical-row extraction is running; the execution log

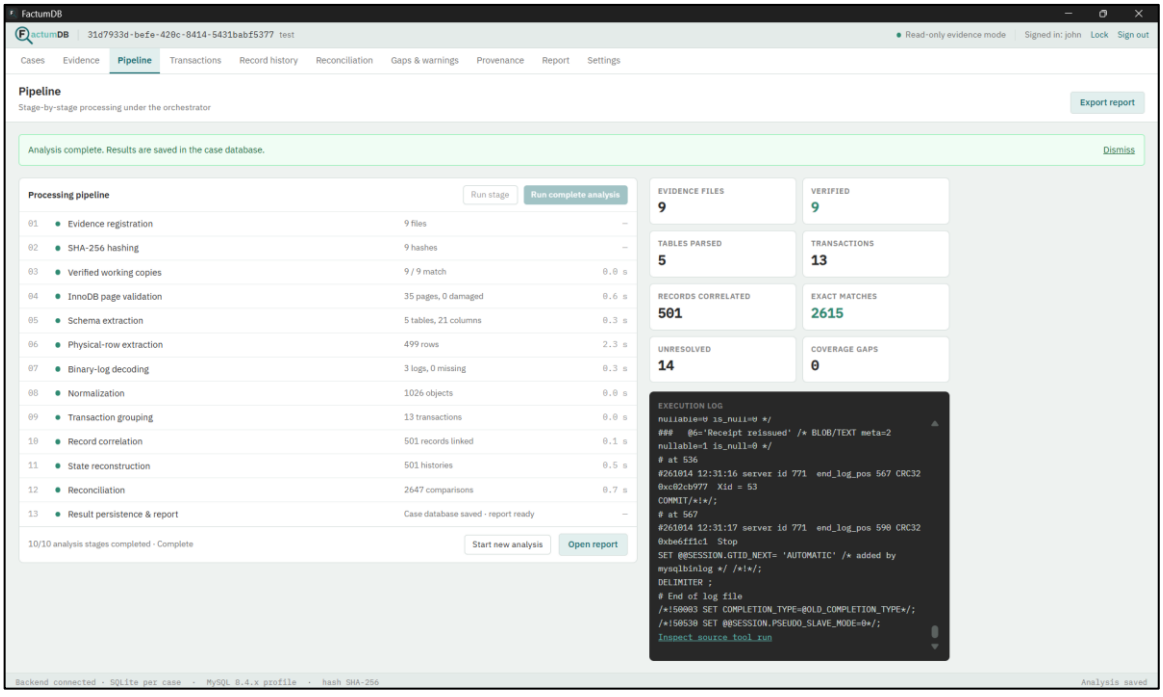


Figure 9. Completed analysis. The banner confirms that results were saved in the case database; summary cards and report controls are now populated.

Completion and example totals

Completion is indicated by the message “Analysis complete. Results are saved in the case database.” The example in Figure 9 shows 9 evidence files, 9 verified copies, 5 parsed tables, 13 transactions, 501 records correlated, 2,615 exact field/presence comparisons, 14 unresolved comparisons, and 0 coverage gaps. These values are demonstration results, not expected values for every case.

After completion, “Open report” opens reporting. “Export report” is also available from analysis-related views.

10. Transactions

The “Transactions” tab groups row events into committed transactions and shows events in commit order. Select a transaction in the left panel to inspect its metadata and row events.

Transaction metadata

- Transaction identifier and commit status.
- GTID when available, XID, source binary log, commit time, start position, end position, and synthesized status.
- Displayed timezone. The example screen explicitly shows UTC; always check the timezone before interpreting chronology.

Row events

- INSERT: a row is introduced. When there is no previous row image, the BEFORE side may state “No row image”.
- UPDATE: compare BEFORE and AFTER values to see which observed fields changed.
- DELETE: the row is removed from the log-derived sequence; available row images depend on the evidence and logging configuration.
- “No row image” means the row image was not present in that event representation. It is not the same as SQL NULL, an empty string, or zero.

Use “Provenance” and “Inspect source tool run” to trace an observation back to extraction details.

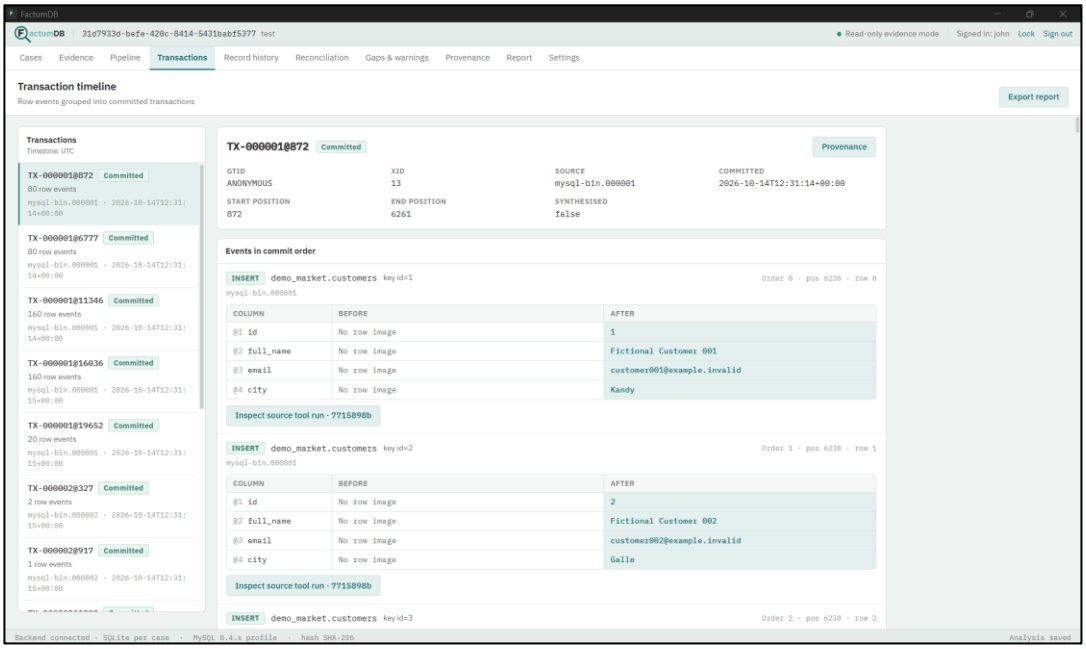


Figure 10. Transaction timeline and row-event details. The selected transaction shows metadata and INSERT events with BEFORE/AFTER values.

11. Record history

1. Open “Record history”.
2. Filter by table when useful, then select a correlated record from the left panel.
3. Read the chronological log-event steps and how each event affects reconstructed committed state.
4. Compare the independent physical-row observation extracted from the .ibd file with the log-derived history.
5. Expand “Observed log row images” and “Findings and reasoning” when additional detail is needed. Use “Open transaction” and “Inspect source tool run” to move to supporting context.

Important distinction A physical row extracted from a .ibd tablespace is an independent observation; it is not a replayed binary-log event. Likewise, the first observed INSERT only marks the earliest INSERT present in the imported evidence. It does not prove that no earlier history existed outside the evidence set.

“Primary-key exact match” describes how the record was correlated. It is separate from the later reconciliation classification such as Exact, Strong, Conflicting, or Unresolved.

The marker “[not observed]” means that FactumDB does not have an observation for that value at that point; it should not be read as NULL or as a known previous value.

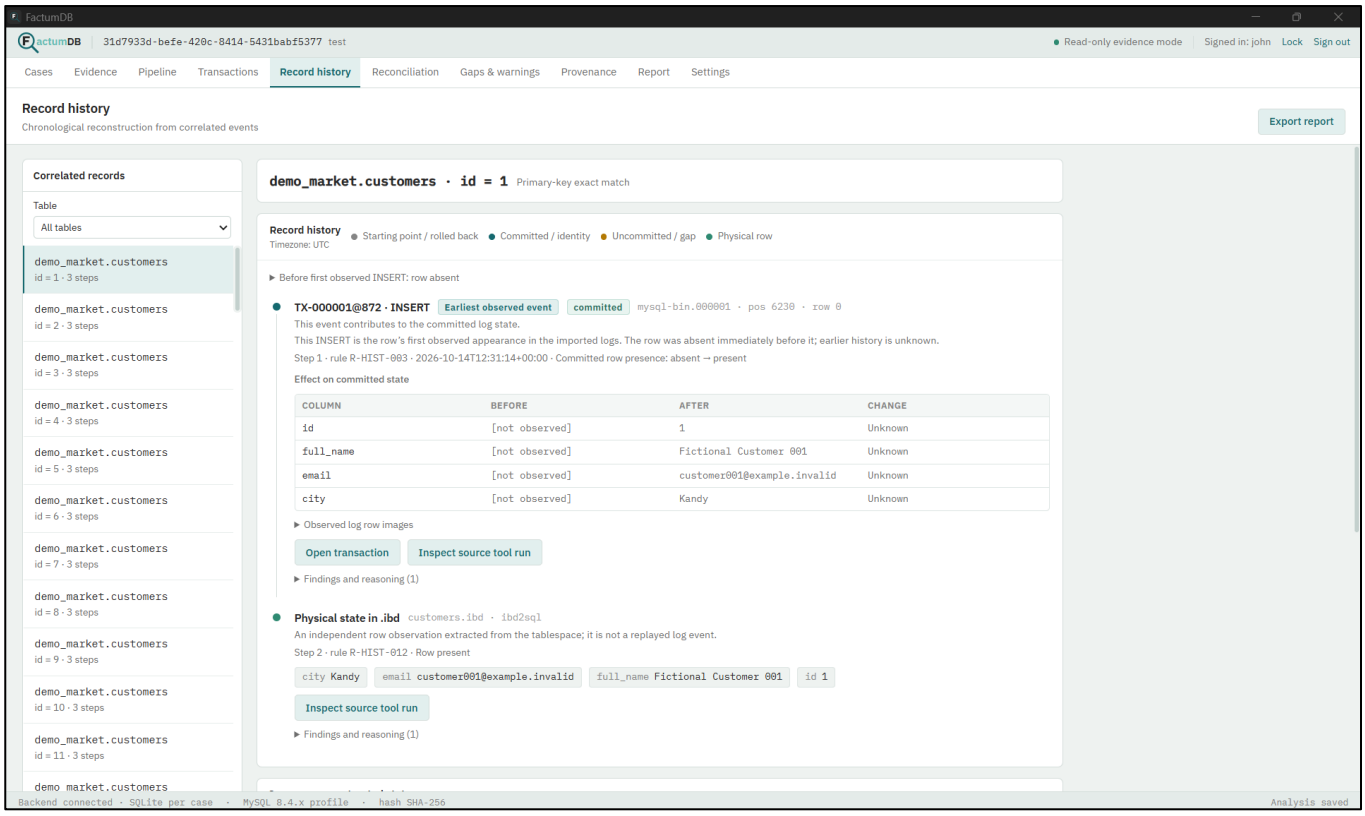


Figure 11. Record history for a correlated customer record. The view separates observed log events, their reconstructed state effect, and an independent physical row from customers.ibd.

12. Reconciliation

Reconciliation compares the log-derived record state with physical .ibd observations under FactumDB's applicable rules. The classification labels describe the result of that comparison; they are not statistical confidence percentages.

Label	How to interpret it
Exact	Compared observations agree under the applicable rules and available evidence coverage.
Strong	Agreement is supported, with evidence limitations or qualifications.
Partial	Only part of the comparison can be established.
Conflicting	Comparable values or record-presence observations disagree.
Unresolved	Evidence is insufficient to reach a conclusion.
Unsupported	The comparison cannot be supported by the available representation or rules.

1. Use "Filter result" to narrow the result set.
2. Expand "View comparisons" for a record to inspect field-level and presence comparisons.
3. Read the log-derived and physical values side by side, together with the result and rule ID.
4. Use "Open record history" when you need the chronological context behind the reconciliation result.

> **Counts use different units** The pipeline example shows 2,615 exact field/presence comparisons, whereas Figure 12 shows 493 records classified as Exact. Do not treat these as competing totals; one counts comparisons and the other counts records.

Figure 12 contains 501 reconciled records: 493 Exact, 2 Strong, 0 Partial, 5 Conflicting, 1 Unresolved, and 0 Unsupported. These are example case results, not expected outcomes for other evidence sets.

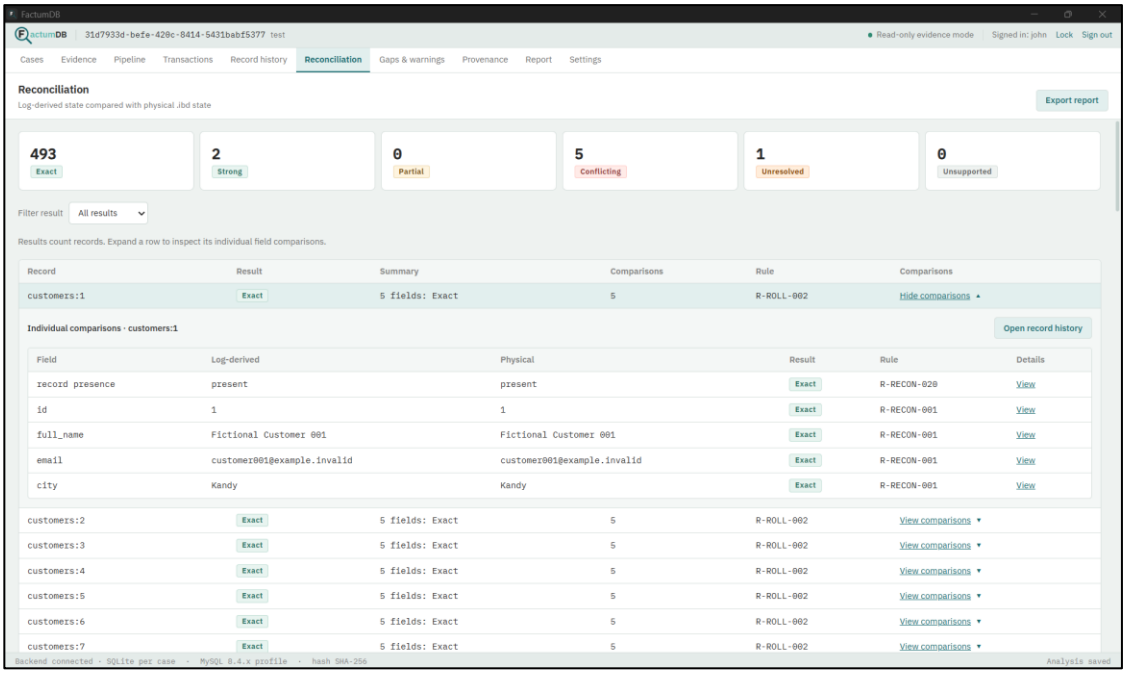


Figure 12. Reconciliation with one record expanded. Field-level log-derived and physical values are compared individually, with result labels and rule IDs.

13. Gaps and warnings

“Gaps & warnings” collects findings that constrain interpretation or require examiner attention. The summary separates warning groups, notice groups, and extraction warning groups.

- Warning groups highlight conditions such as partial row images, disagreements, record-presence conflicts, or insufficient evidence.
- Notice groups provide context that may affect interpretation, such as identity changes, key reuse, physical records without log events, or limited-coverage agreement.
- Extraction warning groups summarize warnings emitted during evidence extraction. A value of zero means no extraction warning groups were stored for the shown run; it does not cancel other warning or notice groups.

1. Review the group summary at the top of the screen.
2. For a rule of interest, select “View findings” to inspect the affected subjects and supporting details.
3. Interpret the finding together with record history, reconciliation, and provenance rather than treating the rule name as a final conclusion.

How counts work Groups are organized by rule. “Occurrences” counts original findings, while “Distinct subjects” counts affected subjects; subjects can overlap across different groups. These counts do not represent separate attacks or separate root causes.

A dashboard value of 0 coverage gaps does not imply that there are no warnings, conflicts, missing observations, or other limitations.

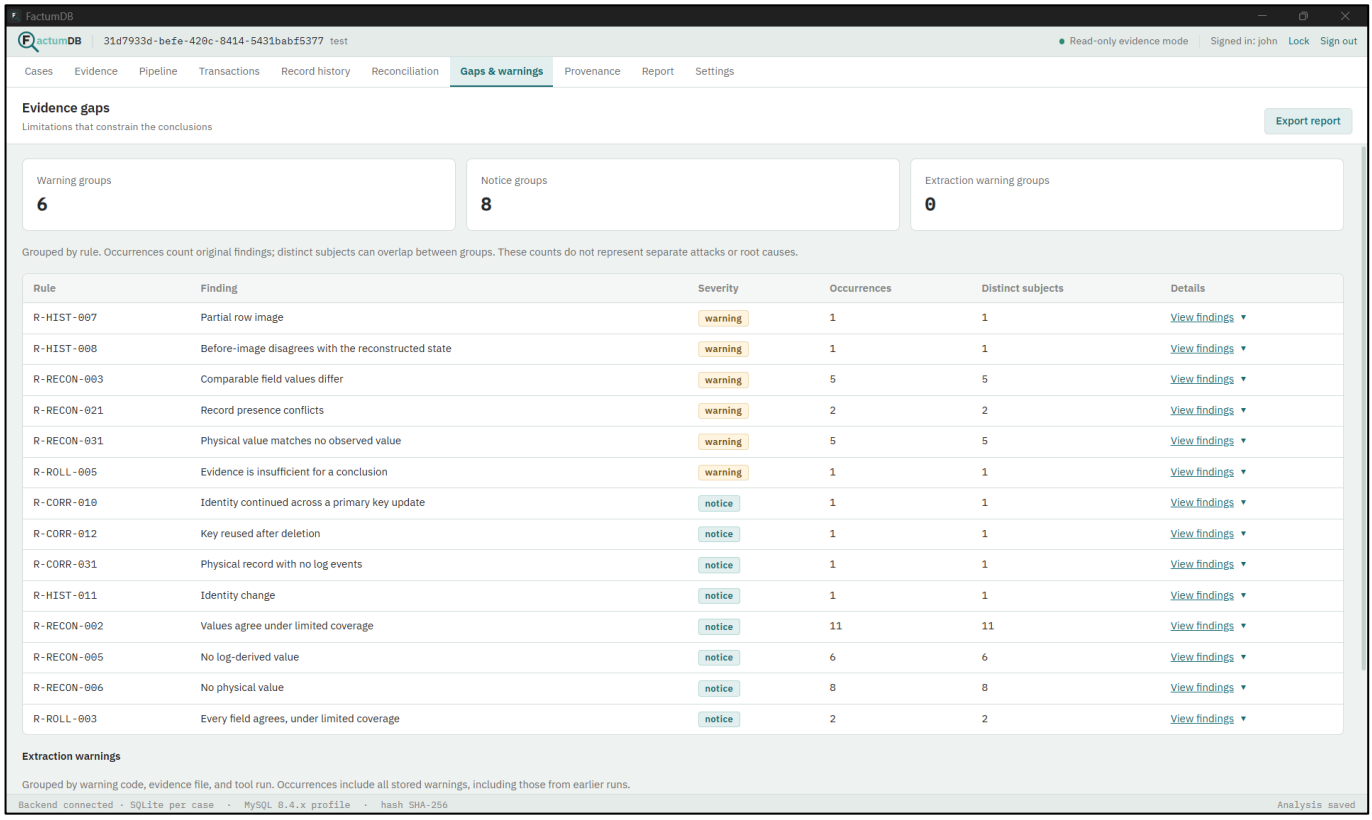


Figure 13. Gaps & warnings. Findings are grouped by rule with severity, occurrences, distinct subjects, and drill-down links.

14. Provenance

Provenance links findings and observations to source evidence and the extraction tool runs that produced supporting outputs. This is central to reproducibility and examiner review.

The table shows utility, version, status, exit code, evidence reference, start time, and an “Inspect run” control. Examples in the application include innochecksum, ibd2sdi, ibd2sql, and mysqlbinlog.

1. Open “Provenance” directly, or follow a “Provenance” / “Inspect source tool run” link from another investigation view.
2. Identify the utility, version, evidence reference, execution status, exit code, and start time.
3. Select “Inspect run” to review stored execution details for that tool invocation.

Tool success is not a forensic verdict Exit code 0 and “succeeded” indicate that the extraction command completed successfully. They do not prove that the evidence was complete, that every record was recoverable, or that a downstream interpretation is correct.

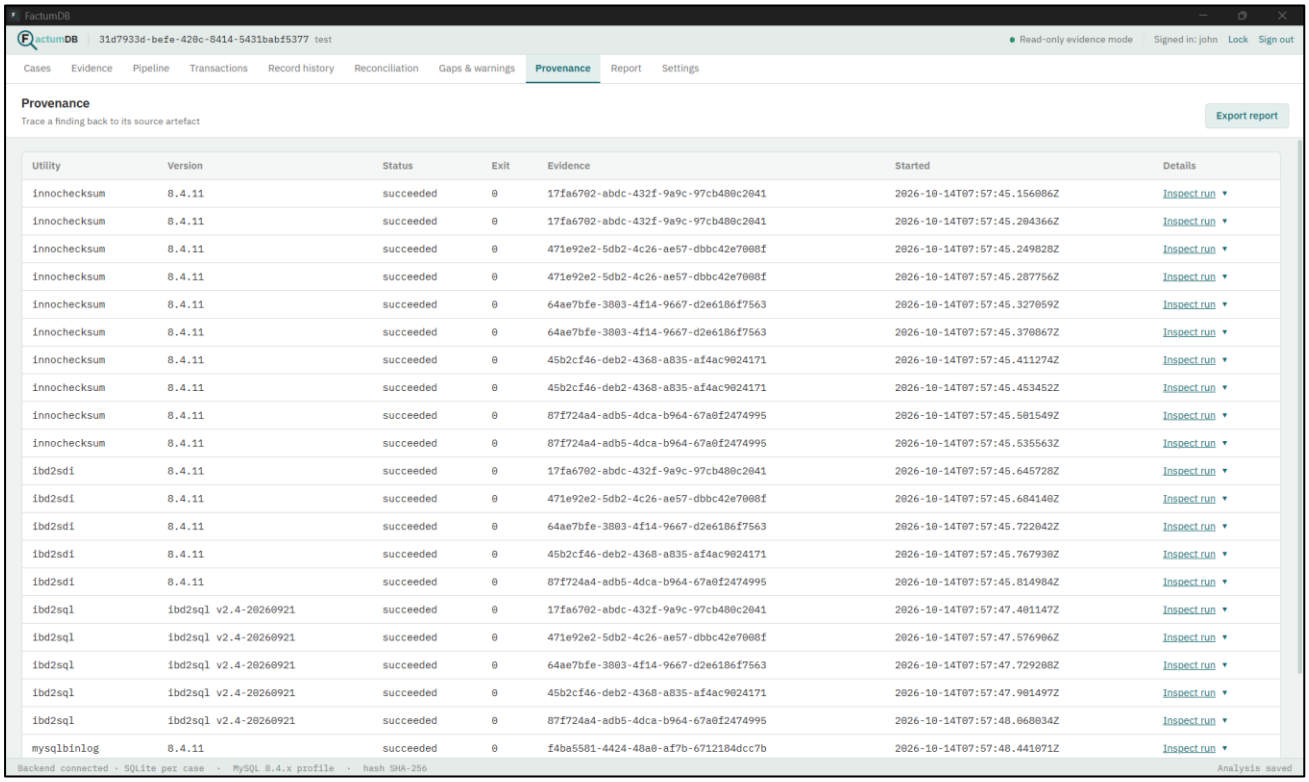


Figure 14. Provenance table showing extraction utilities, versions, execution status, evidence references, times, and “Inspect run” links.

15. Report export

FactumDB provides two report formats. JSON is a detailed full-case export. PDF is an investigator summary that includes evidence information, classification totals, and grouped findings; it does not include every individual record.

Content modes

- “Redacted sharing copy”: replaces database identifiers, database values, and evidence filenames with aliases. It retains case names, examiner identity, tool metadata, provenance IDs, hashes, timestamps, and acquisition methods. Notes, paths, commands, and diagnostics are withheld.
- “Original values (sensitive)”: retains sensitive original content appropriate to the selected format. A PDF remains a summary even when original values are selected.

Redacted does not mean anonymous The sharing copy is pseudonymized and does not guarantee anonymity. It is unsuitable for independent reanalysis. The original case data remains intact inside FactumDB.

Export workflow

1. Open “Report”.
2. Select JSON or PDF.
3. Choose the content mode.
4. Review the preview or select “Refresh preview”.
5. For PDF, use Previous/Next and Zoom to inspect pages. Review coverage counts where they are available.
6. Select “Choose destination” and specify a new output filename.
7. Select the format-specific export or generation button, such as “Generate PDF report”.
8. Check the saved confirmation and export history.

A fresh preview and valid destination are required. Existing exports are never overwritten. If an export control is disabled, check analysis completion, preview readiness, and whether a new destination has been selected.

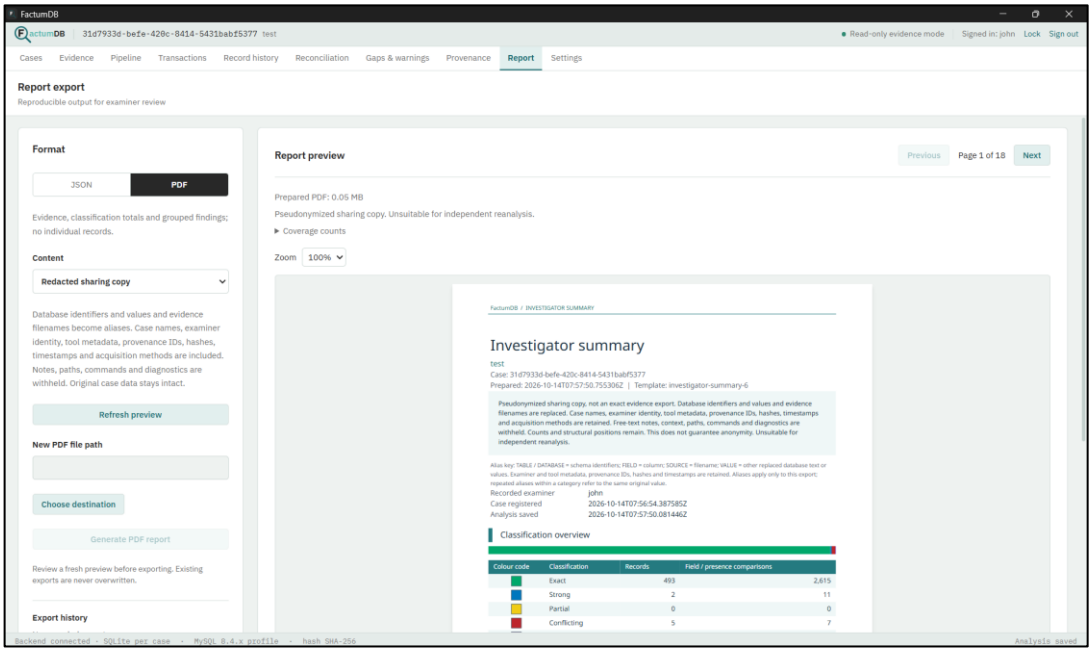


Figure 15. FactumDB PDF report preview and export controls. The previewed investigator summary is an application-generated case report, not this user guide.

16. Settings and troubleshooting

Settings

Packaged builds automatically select bundled extraction tools. If paths have been overridden, “Use bundled tools” restores the packaged tool paths. Settings also includes profile, password, preferences, and workspace backup/restore controls. This guide does not prescribe a detailed backup/restore procedure because the interface is not shown in the supplied screenshots.

Troubleshooting

Problem	What to check	Safe next action
Windows sign-up verification unavailable	Windows Hello is configured for the current Windows user.	Configure or repair Windows Hello, then retry sign-up.
Ubuntu verification unavailable	Active graphical session, polkit authentication agent, and installed FactumDB polkit policy.	Restore the desktop authentication path, then retry. Do not bypass verification.
Evidence remains Registered	“Register & verify” was run and no verification error is pending.	Run verification and inspect any displayed error before analysis.
Tool execution failure	Execution log, provenance, and bundled-tool settings.	Inspect the failed tool run; use bundled tools if a custom override is invalid.
Empty results	Correct case, imported evidence, completed stages, and evidence coverage.	Confirm the selected case and evidence, then review pipeline and warnings.
Export disabled or stale	Analysis completion, current preview, and output destination.	Refresh preview and choose a new destination.
Existing export filename	FactumDB does not overwrite prior exports.	Choose a different output filename.

Preserve evidence and security controls Do not modify original evidence to make processing succeed. Do not disable operating-system security controls or bypass native authentication. Diagnose the working-copy, tool, or environment issue instead.

17. Glossary and interpretation checklist

Glossary

Term	Meaning in this guide
.ibd	An InnoDB tablespace file that can contain physical database pages and row records.
Binary log	A MySQL log of database changes over time. FactumDB decodes row events from the imported log sequence.
Working copy	The verified copy that FactumDB processes while preserving the original evidence.
SHA-256	A cryptographic hash used here to compare registered evidence and working copies for integrity verification.
Transaction	A committed group of row events shown in commit order, with available metadata such as XID and source log positions.
Reconstruction	The process of applying observed log events to derive the record state represented by those events.
Reconciliation	Comparison of log-derived state with physical .ibd observations under explicit rules.
Provenance	Information that links an observation or finding to evidence and the tool run that produced supporting output.

Before you state a conclusion

- Separate what was observed from what is inferred.
- Check whether the relevant binary-log interval and tablespace evidence are actually available.
- Treat “No row image” and “[not observed]” as absence of an observation, not as a database value.
- Distinguish record classifications from field/presence comparison counts.
- Review Gaps & warnings for evidence limitations and identity/key edge cases.
- Use Provenance to identify the source evidence and extraction run behind important findings.
- Do not equate Exact with authenticity, Conflicting with malicious activity, or successful processing with completeness.

Project reference

Repository: https://github.com/Factum-Labs/FactumDB